Business Question and Visualization Report

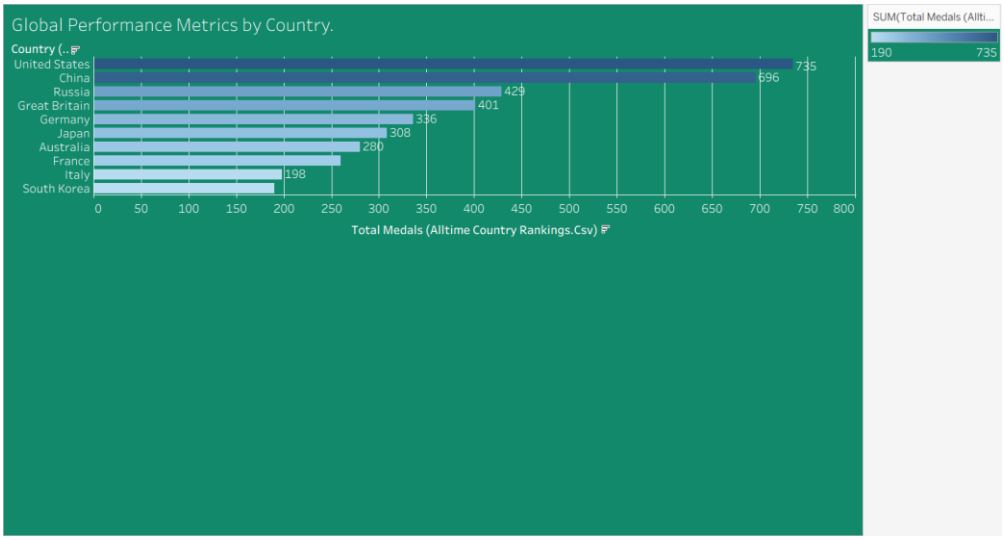
Date	28 October 2026
Team ID	EN25CS3010040
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

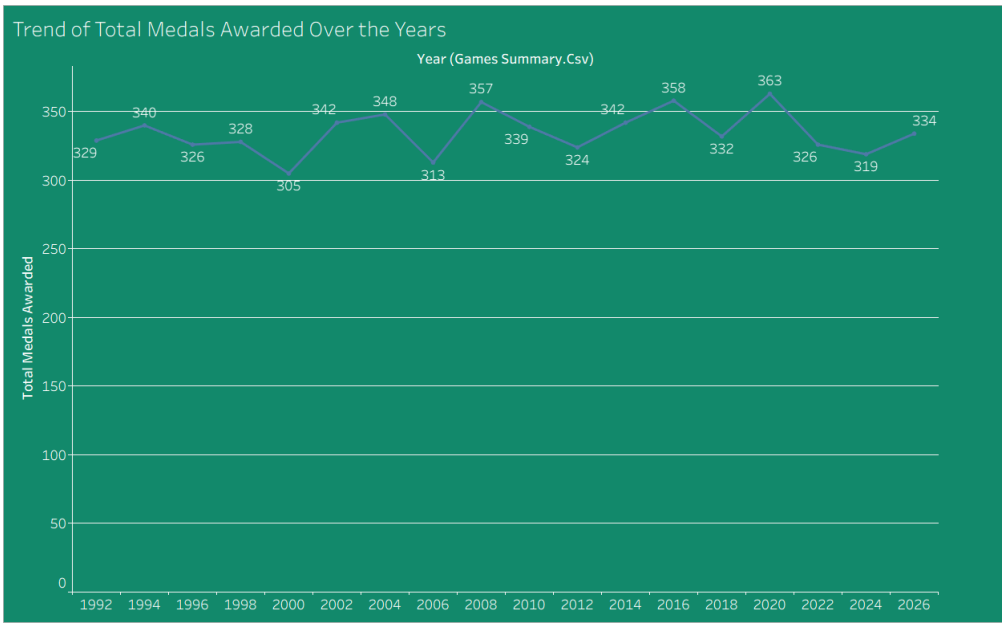
The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. **Which countries have the highest total Olympic medal counts?**
    - *Visualization*: Horizontal bar chart showing total medals for the top 10 countries from alltime_country_rankings.csv, sorted in descending order. The United States leads with 735 medals, followed by China (696) and Russia (429).
    - *Screenshot of visualisation*



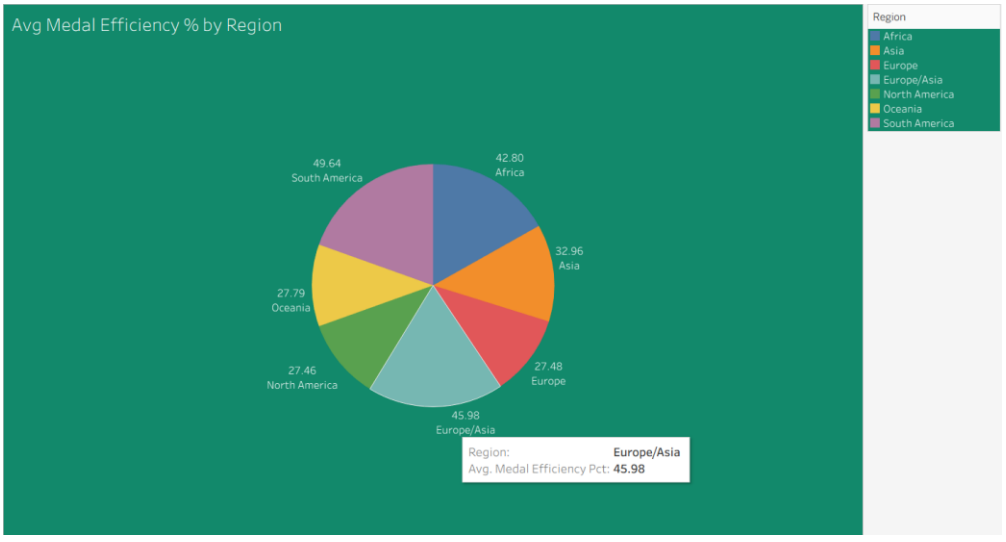
2. How have Olympic medal counts changed over the years?

- *Visualization*: Line chart showing total medals awarded by year from 1992 to 2026 using games_summary.csv. Totals stay between 305 (2000) and 363 (2020), with no steady rise or fall.
- *Screenshot of visualisation*



3. How does average medal efficiency vary by region?

- *Visualization*: Pie chart showing average medal efficiency percentage by region. South America is highest (49.64), followed by Europe/Asia (45.98) and Africa (42.80); North America is lowest (27.46).
- *Screenshot of visualisation*

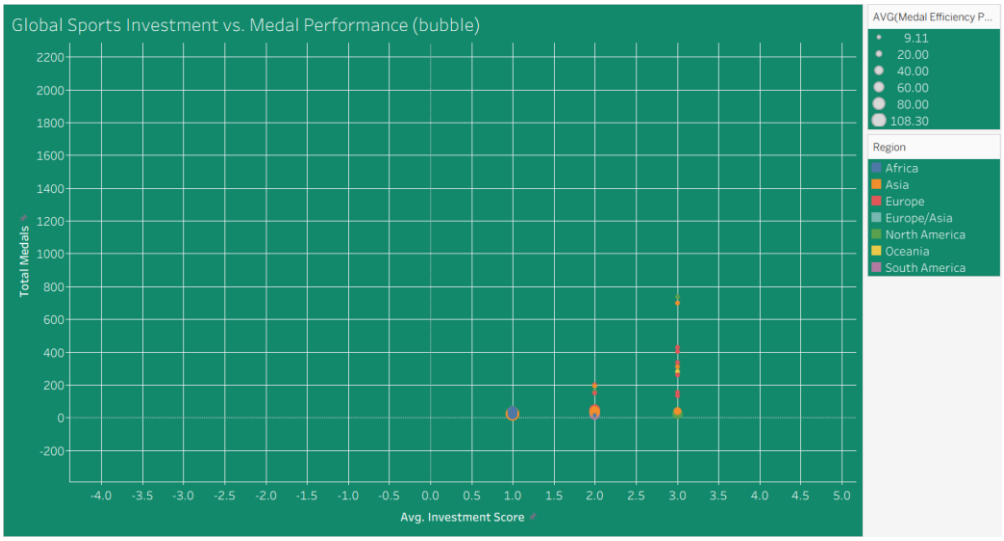


4. How does sports investment relate to medal performance?

- *Visualization*: Bubble chart plotting average investment score against total medals, with bubble size showing average medal efficiency and colour

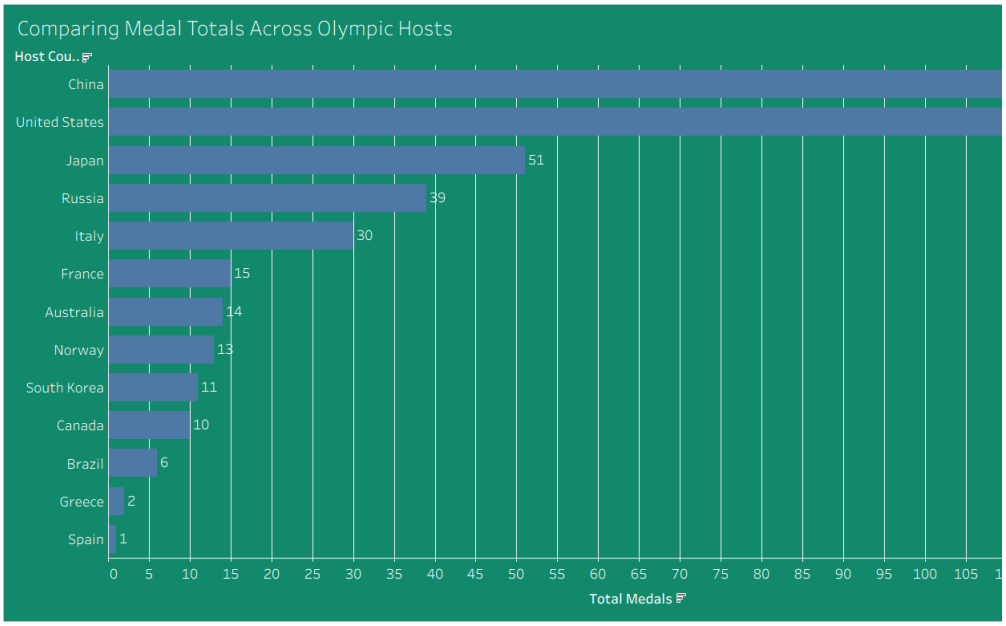
showing region. The countries with the highest medal totals sit at the top investment score of 3.

- *Screenshot of visualisation*



5. How does medal performance vary by host country?

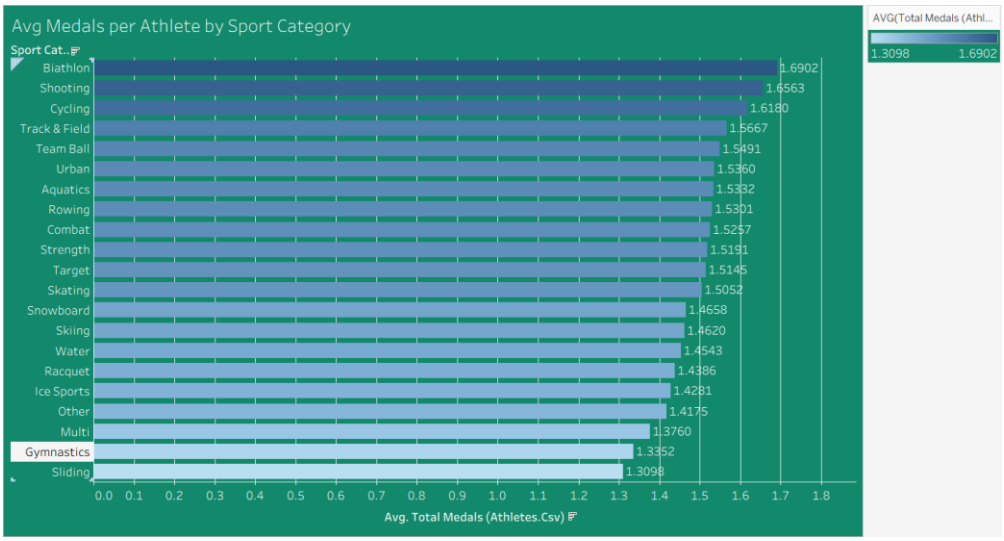
- *Visualization*: Bar chart comparing total medals across Olympic host countries (filtered to host nations). China (118) and the United States (116) lead, followed by Japan (51), Russia (39) and Italy (30).
- *Screenshot of visualisation*



6. Which sport categories achieve the highest average medals per athlete?

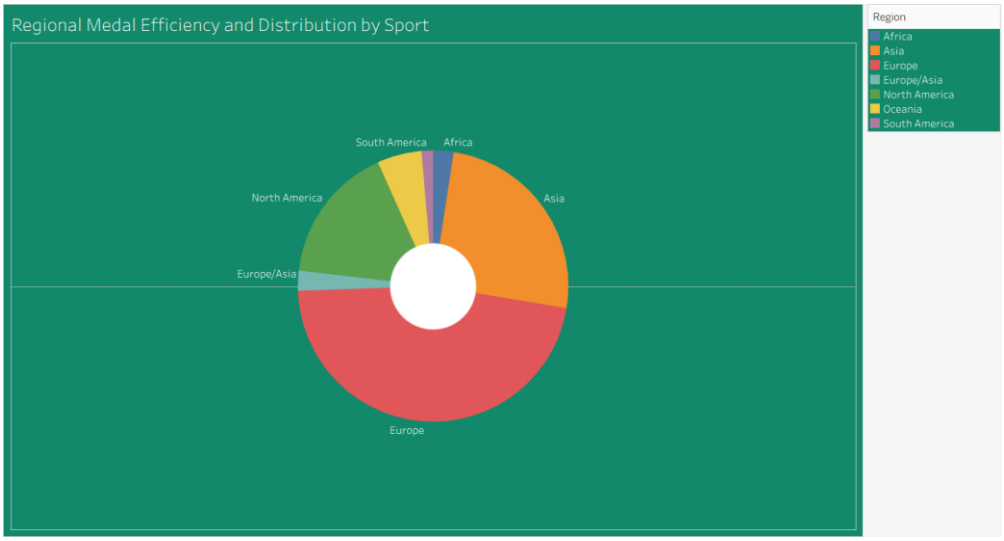
- *Visualization*: Horizontal bar chart showing average medals per athlete by sport category using athletes.csv, with a colour gradient from 1.3098 to 1.6902. Biathlon (1.6902), Shooting (1.6563) and Cycling (1.6180) rank highest; Sliding (1.3098) ranks lowest.

- *Screenshot of visualisation*



7. How is medal efficiency distributed across regions?

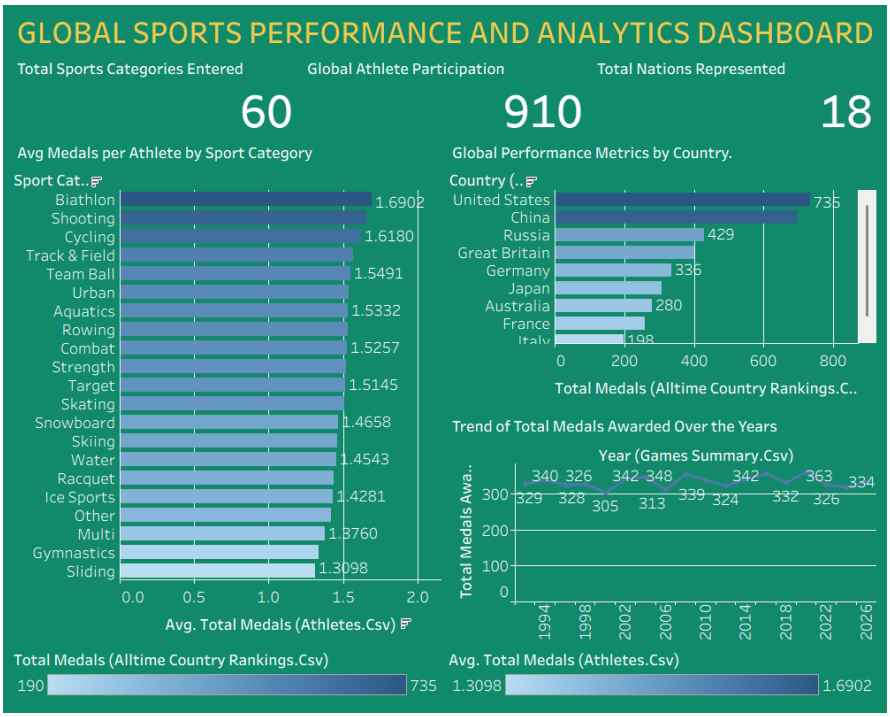
- *Visualization*: Donut chart showing the share of each region, colour-coded by region. Europe holds the largest share, followed by Asia, with North America, Oceania, Europe/Asia, South America and Africa making up the rest.
- *Screenshot of visualisation*



8. How do the key performance indicators and charts come together in the dashboards?

- *Visualization*: Two interactive dashboards combining the worksheets. Dashboard 1 (Global Sports Performance and Analytics) shows 60 sport categories, 910 athletes and 18 nations alongside the sport, country and yearly trend charts. Dashboard 2 (Olympic Performance and Regional Efficiency) shows 910 athletes and a regional performance efficiency of 31.8 with the region, investment, host and distribution charts.

- *Screenshot of visualisation (Dashboard 1)*



- *Screenshot of visualisation (Dashboard 2)*

